

```
GENERICSEntry top;class Box<T> {}// variableBox<String> b = new Box<String>();// methodpublic <T> void doStuff(T value) {}
```

```
JAVA BEING STUPIDT obj = new T(); // errorobj instanceof T; // errorT heh = (T) "asdf"; // no typecheckingvar s = Stack<int>(); // no primitive typesprivate static T x; // static no workeyNode<Number> s = new Node<Integer>; // errorclass IThrowAnErrorBecauseWhyNot {void m(List<String> list) {}void m(List<Integer> list) {}// error because of type erasure & identifiabilityInteger[] c = set.toArray(new Integer[0]);}
```

```
ITERATORfor (String s : stringList) {} // ==for (Iterator<String> i = stringList.iterator(); i.hasNext();) {}String s = i.next();}
```

```
ITERABLEinterface Iterable<T> {Iterator<T> iterator();}
```

```
interface Iterator<E> {boolean hasNext();E next();}
```

```
COMPARABLETODO:Type signature?
```

```
static <T extends Comparable> T doStuff(T a, T b) { // compareTo is possible with all typesreturn a.compareTo("b") > 0 ? a : b;}
```

```
static <T extends Comparable<T>> T doStuff(T a, T b) { // compareTo is possible only with Treturn a.compareTo(b) > 0 ? a : b;}
```

```
EXAMPLEclass Graphic {public void draw() {}class Rectangle extends Graphic {}class Circle extends Graphic {}class GraphicStack<T extends Graphic> extends Stack<T> {public void drawAll() {for (T item : this) item.draw();}}
```

```
BYTECODETODO:
```

```
TYPE ERASURE– Introduced because of “bAcKwArDs CoMpAtAbIlITy”– No type information at runtime (Non-Reifiable Type)
```

```
class MyStack<T> {Entry<T> top;int push(T value) {}// becomesclass MyStack {Entry top;int push(Object value) {}/* with bounds */class MyStack<T extends Comparable<T>> {Entry<T> top;int push(T value) {}// becomes
```

```
class MyStack {Entry top;int push(Comparable value) {}// what happens at/before runtime */MyStack<String> s = new MyStack<String>();s.push("Eh");String x = s.pop();// becomesMyStack s = new MyStack();s.push("Eh");String x = (String) s.pop();
```

```
WILDCARDSvoid printGs(List<Graphic> gs) {}List<Graphic> gs1 = new ArrayList<>();printGs(gs1); // okList<Circle> gs2 = new ArrayList<>();printGs(gs2); // error/* solution */void printGs(List<? extends Graphic> gs) {}
```

|                | Type           | Compatible Type-Args | R | W |
|----------------|----------------|----------------------|---|---|
| Invariance     | C<T>           | T                    | ✓ | ✓ |
| Covariance     | C<? extends T> | T and sub-types      | ✓ | × |
| Contravariance | C<? super T>   | T and basis-types    | × | ✓ |
| Bivariance     | C<?>           | All                  | × | × |

```
INVARIANCEstatic <T> void move(Stack<T>, from, Stack<T> to) {while(!from.isEmpty()) to.push(from.pop());}var rectS = new Stack<Rectangle>();var graphS = new Stack<Graphic>();graphS.push(new Rectangle()); // okmove(rectS, graphS); // errorStack<Object> objS = graphS; // error/* anotherone */String[] strA = new String[10];Object[] objA = strA; // okobjA[0] = Integer.valueOf(2); // runtime err
```

```
TODO:// compiles?List<Graphic> list = new ArrayList<Rectangle>();// compiles?Rectangle[] rectangleStack = new Rectangle[10];Graphic[] graphicStack = new Graphic[10];graphicStack = rectangleStack; // compiles?Stack<Rectangle> rectangleStack = new Stack<>();Stack<Graphic> graphicStack = new Stack<>();graphicStack = rectangleStack; // compiles?Stack<Rectangle> rectangleStack = new Stack<>();Stack<Graphic> graphicStack = new Stack<>();for (Rectangle r : rectangleStack) {graphicStack.push(r);}
```

```
COVARIANCEpublic class Stack<E> {public void pushAll(Iterable<? extends E> src) {}Stack<? extends Graphic> graphS;graphS = new Stack<Rectangle>(); // okgraphS = new Stack<Circle>(); // okgraphS = new Stack<Object>(); // error (duh)/* read only */Stack<? extends Graphic> stack = new Stack<Rectangle>();stack.push(new Graphic()); // errorstack.push(new Rectangle()); // errorstack.push(new Triangle()); // errorCONTRAVARIANCEstatic <T> void addToC(List<? super T> list, T e) {list.add(e);}addToC(new ArrayList<Number>(), 3); // okaddToC(new ArrayList<Object>(), 3); // ok
```

```
/* shenanigans */Stack<? super Graphic> s = new Stack<Object>();s.add(new Graphic()); // oks.add(new Rectangle()); // okGraphic g = s.pop(); // errorObject g = s.pop(); // ok
```

```
BIVARIANCEstatic void printList(List<?> list) {for (Object elem : list)System.out.print(elem + " ");}/* more shenanigans */static void appendNewObject(List<?> list) {list.add(new Object()); // error}
```

```
PRODUCER / CONSUMERTODO:
```

```
MISC<T extends Comparable & Collection> // multiple type boundsGENERIC STREAM TOMFOOLERYList<? extends Media> medialis;public <S extends T> List<S>filterBySubtype(Class<S> subtype) {return medialis.stream().filter(subtype::isInstance).map(subtype::cast).collect(Collectors.toList());}
```

```
ANNOTATIONS– Metadata– Not actually part of the code
```

```
TODO:slides 10
```

```
– @Override– @Test– @Json...
```

```
DEFINING@Target(ElementType.TYPE)@Retention(RetentionPolicy.RUNTIME)public @interface Author {String name();String date() default "N/A";}
```

```
@Author(name = "UwU", date = "idon't validate input")public class WeeDoo {}
```

```
REFLECTON– Information of Class (private and public): Methods, Attribs, Parent class, Implemented interfaces– Calling methods possible– Usages: Debugger, Serialization, Frameworks, Remote Procedure Call, ORM
```

```
// all of these `throws SecurityException`public Field[] getDeclaredFields()public Method[] getDeclaredMethods()public Constructor<?>[] getDeclaredConstructors()
```

```
System.out.println(dump(new Student("a", "b"), 0));// { // firstName = a // lastName = b // }Class c = "s".getClass(); // java.lang.String
```

```
TODO:Diagram (slides 26)
```

```
METHOD// @Target(ElementType.METHOD)public String getName()public boolean isAnnotationPresent(Class<? extends Annotation> annotationClass)public Object invoke(Object obj, Object... args)
```

```
throws IllegalAccessException, IllegalArgumentException, InvocationTargetException
```

```
public class Profiler {public static void main(String[] args) {var testFunctions = new ProfileFunctions();int[] array = {5, 2, 8, 1, 93, 3, 33, 1, 333};var methods = ProfileFunctions.class.getDeclaredMethods();for (var m : methods) {if(m.isAnnotationPresent(Profile.class)) {m.setAccessible(true); // best practice orsth, idk, i write code in actually good languagesProfiler.profileMethod(testFunctions, m, new Object[] {array});}}
```

```
CLASSTODO:
```

```
// @Target(ElementType.TYPE)getDeclaredConstructor().newInstance()...
```

```
ATTRIBUTETODO:
```

```
// @Target(ElementType.FIELD)
```

```
VALIDATION
```

```
@Min(value = 18, message = "Age must be ≥ 18")@Max(value = 99, message = "Age must be ≤ 99")private int age;@NotNull(message = "Name cannot be null")private String name;
```

```
ALGORITHMS
```

- Brute-force
- Greedy
  - Take best option each step (optimize locally)
  - Fast
- Backtracking
  - Trial and error
  - Best overall option
  - Higher compute costs
- Divide and conquer
  - Binary search
- Dynamic programming
  - Recursion optimization?
  - Tabulation

```
BACKTRACKINGTODO:slides 87
```

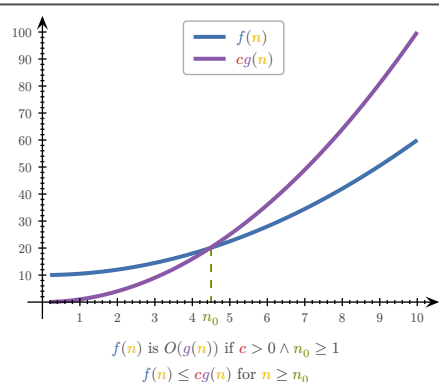
```
BIG O NOTATION– Worst case scenario– Atomic operations = constant time– Runtime measured as sum of primitive operations
```

Notation

$$\widetilde{f} \left( \underbrace{n}_{\text{Size of problem}} \right) = \text{Number of steps}$$

$g = \text{complexity class}$

```
TODO:diagram  $f \leq cg$ 
```



$f$  in order  $g$ :

Let  $f, g : \mathbb{N} \rightarrow \mathbb{N}$

$n_0, c \in \mathbb{N} \wedge \forall n \geq n_0 : f(n) \leq c \cdot g(n)$

$\Rightarrow f \in O(g) \Leftrightarrow f = O(g)$

Big O

$$\overset{\text{Big O}}{\underset{\text{Size of problem}}{\underbrace{n}}} = \text{Complexity class}$$

RULES

If  $f(n)$  is polynomial of degree  $d$ , then  $f(n) \in O(n^d)$

- ignore lower powers
- ignore constants

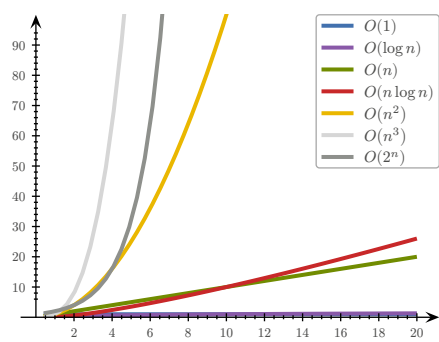
Use most optimal function (lowest power)

$2n$  is  $O(n)$  better than  $2n$  is  $O(n^2)$

Simplify as far as possible

$3n + 5$  is  $O(n)$  instead of  $3n + 5$  is  $O(3n)$

| COMPLEXITY CLASSES |            |                            |
|--------------------|------------|----------------------------|
| Name               | Class      | Example                    |
| Constant           | 1          | Array read                 |
| Logarithmic        | $\log n$   | Binary search              |
| Linear             | $n$        | Linear search              |
| Log-linear         | $n \log n$ | Merge+Quick sort           |
| Quadratic          | $n^2$      | Bubble+Insertion sort      |
| Cubic              | $n^3$      | Solve equation             |
| Polynomial         | $n^k$      | Various algorithms         |
| Exponential        | $2^n$      | Calculate all subsets      |
| Factorial          | $n!$       | Calculate all permutations |



```
TODO:Counting primitive operations (slides 55)
```

SORTING ALGORITHMS

SELECTION SORT

- Search for smallest elem in unsorted part and move in front

- Less mutations in array
- Same performance even if list almost sorted

```
public static void selectionSort(int[] a) {
    int n = a.length;
    for (int i = 0; i < n - 1; i++) {
        int minimum = i;
        for (int j = i + 1; j < n; j++)
            if (a[j] < a[minimum])
                minimum = j;
        swap(a, i, minimum);
    }
}
```

$$(n-1) + (n-2) + \dots + 1 = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \frac{n^2-n}{2} \sim O(n^2)$$

- INSERTION SORT**
- Take elem and put into correct place in sorted part
  - Good for already partially sorted arrays

```
public static void insertionSort(int[] a) {
    int n = a.length;
    for (int i = 1; i < n; i++)
        for (int j = i; j > 0 && a[j] < a[j - 1]; j--)
            swap(a, j, j - 1);
}
```

$$1 + \dots + (n-1) + n = \sum_{i=1}^n i = \frac{n(n+1)}{2} = \frac{n^2+n}{2} \sim O(n^2)$$

- BUBBLE SORT**
- Compare neighboring elems and swap if needed

```
public static void bubbleSort(int[] a) {
    for (n = a.length; n > 1; n--)
        for (i = 0; i < n - 1; i++)
            if (a[i] > a[i + 1])
                swap(a, i, i + 1);
}
```

$$(n-1) + (n-2) + \dots + 1 = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \frac{n^2-n}{2} \sim O(n^2)$$

```
public static void countingSort(int[] a) {
    int k = Arrays.stream(arr).max().getAsInt();
    int[] c = new int[k + 1];
    for (int i : a) c[i] += 1;
    int pos = 0;
    for (int i = 0; i < k; i++)
        for (int j = 0; j < c[i]; j++) {
            a[pos] = i;
            pos++;
        }
}
```

$$n + n + k + n = 3n + k \sim O(n + k)$$

- SHELL SORT**
- Sort pairs of far away elems
  - Sort the partially sorted array through insertion sort

```
public static void shellSort(int[] a) {
    int n = a.length;
    int h = 1;
    while (h < n/3) h = 3 * h + 1;
    while (h >= 1) {
        for (int i = h; i < n; i++)
            for (int j = i; j >= h && a[j] < a[j - h]; j = j - h) swap(a, j, j - h);
        h /= 3;
    }
}
```

**BINARY SEARCH**

```
public static <T extends Comparable<T>> boolean bs(
    List<T> data, T target, int low, int high) {
    if (low > high) return false;
    int pivot = low + ((high - low) / 2);
    if (target.equals(data.get(pivot)))
        return true;
    else if (target.compareTo(data.get(pivot)) < 0)
        return searchBinary(data, target, low, pivot - 1);
    else
        return searchBinary(data, target, pivot + 1, high);
}
```

```
return searchBinary(data, target, pivot + 1, high);
}
```

$$O(\log n)$$

- RECURSION**
- When a function calls itself
- LINEAR RECURSION**
- Recursive call starts **at most one** further recursive call

**TODO:**  
recursive → iterative (slides 61)

- TAIL RECURSION**
- Linear recursion where the recursive call is **last**

**TODO:**  
recursive → iterative (slides 69 – nice)

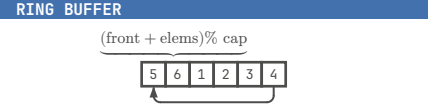
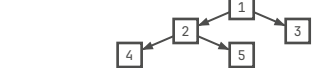
- BINARY RECURSION**
- All non-terminal calls have **two** recursive calls

**TODO:**  
example (slides 76,77,78)

**TODO:**  
 $O(n)$  (slides 84)

- DATA STRUCTURES**
- ADT: Abstract Data Type (e.g. Interface)

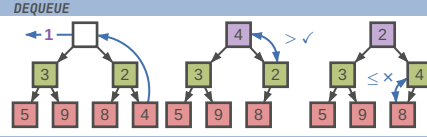
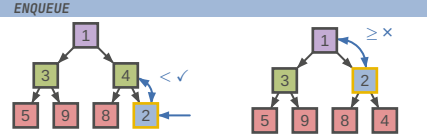
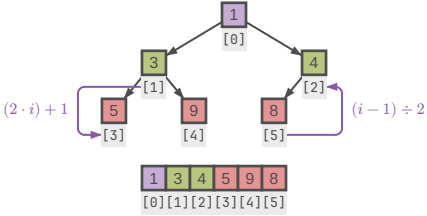
**TODO:**  
diagram



| Method    | Unsorted List | Sorted List |
|-----------|---------------|-------------|
| insert    | $O(1)$        | $O(n)$      |
| min       | $O(n)$        | $O(1)$      |
| removeMin | $O(n)$        | $O(1)$      |

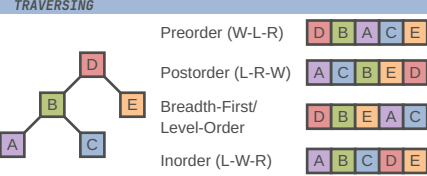
- HEAP**
- Binary tree
  - Layers 0, ...,  $h-1$  fully populated, layer  $h$  filled from left
  - Operations always lowest RHS so that tree stays consistent

- Min-Heap:** Keys of nodes are **smaller** or equal than children's
- Max-Heap:** Keys of nodes are **larger** or equal than children's



- HEAP SORT**
- Dequeue elems from head into list, resorting tree on each iter

- TODO:**
- BINARY SEARCH TREE (BST)**
- All subnodes of **left** subtree are **smaller** than root
  - All subnodes of **right** subtree are **larger** than root



- TODO:**
- MORE DATA STRUCTURES**
- Set:** Elements of same type, no duplicates, without order
  - Multiset:** Set with duplicates
  - Map:** KV-pairs with unique keys
  - Multimap:** Key can have multiple values

- HASHING**
- Hash-function  $h$  maps  $e$  onto  $[0, n-1]$ .  $h(e)$  = position of  $e$

$$e \xrightarrow{\text{hash code}} \mathbb{Z} \xrightarrow{\text{compression function}} [0, n-1]$$

- Properties of good hash functions:
- Equal distribution
  - Fast computation
  - Constant complexity
  - Deterministic
  - Includes as much information from key as possible

- HASH FUNCTIONS**
- Modulo primes
  - Memory address (Object.hashCode() default)
  - Byte/Integer cast ByteBuffer.wrap(b).getInt()
  - Component sum (eg. sum string char codepoints)
  - Polynome accumulation  $a_0 + a_1z + a_2z^2 + \dots + a_{n-1}z^{n-1}$

- ANOMALIES**
- Empty spaces
  - Collisions
  - Overflow

- MANAGING HASH COLLISIONS**
- Closed addressing/Separate chaining:** Each bucket is a list. Search:  $O(\text{entries/buckets})$
  - Open addressing:** "overflows" into next empty bucket

- EQUALITY**
- It is generally necessary to override hashCode whenever equals is overridden (equal objects = equal hashCodes)